

```

1  clc;
2  clear;
3  close all;
4
5  % Input values
6  x = -10:0.1:10;
7
8  % Weight and Bias
9  w = 1;
10 b = 0;
11
12 % Weighted Sum
13 net = w*x + b;
14
15 % Sigmoid Activation Function
16 y = 1 ./ (1 + exp(-net));
17
18 % Display sample values
19 disp(' Input      Output');
20 disp([x(1:10:end)' y(1:10:end)']);
21
22 % Plot
23 figure;
24 plot(x,y,'b','LineWidth',2);
25 grid on;
26 xlabel('Input');
27 ylabel('Neuron Output');
28 title('Artificial Neuron using Sigmoid Activation');

```

Input	Output
-10.0000	0.0000
-9.0000	0.0001
-8.0000	0.0003
-7.0000	0.0009
-6.0000	0.0025
-5.0000	0.0067
-4.0000	0.0180
-3.0000	0.0474
-2.0000	0.1192
-1.0000	0.2689
0	0.5000
1.0000	0.7311
2.0000	0.8808
3.0000	0.9526
4.0000	0.9820
5.0000	0.9933
6.0000	0.9975
7.0000	0.9991
8.0000	0.9997
9.0000	0.9999
10.0000	1.0000

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